

①

$$\int_1^{+\infty} \frac{dx}{x^2(x+1)} = \int_1^{+\infty} \frac{dx}{1} \cdot \left(\frac{A}{x} + \frac{B}{x^2} + \frac{C}{x+1} \right) = \int_1^{+\infty} \left(\frac{-1}{x} + \frac{1}{x^2} + \frac{1}{x+1} \right) dx \quad \text{②}$$

$$\frac{1}{x^2(x+1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x+1} \quad | \cdot (x^2)(x+1)$$

$$1 = Ax(x+1) + B(x+1) + C(x^2)$$

$$Ax^2 + Ax + Bx + B + Cx^2 = 1$$

$$\begin{cases} A+C=0 \\ A+B=0 \\ B=1 \end{cases} \Rightarrow \begin{cases} A=-1 \\ B=1 \\ C=1 \end{cases} \Rightarrow \frac{1}{x^2(x+1)} = \frac{-1}{x} + \frac{1}{x^2} + \frac{1}{x+1}$$

$$\textcircled{2} \lim_{b \rightarrow +\infty} \int_1^b \left(\frac{1}{x+1} - \frac{1}{x} + \frac{1}{x^2} \right) dx = \lim_{b \rightarrow +\infty} \left(\ln(x+1) - \ln x + \frac{x^{-2+1}}{-2+1} \right) \Big|_1^b =$$

$$= \lim_{b \rightarrow +\infty} \left(\ln(b+1) - \ln b - \frac{1}{b} - \left(\ln(1+1) - \ln 1 - \frac{1}{1} \right) \right) =$$

$$= \lim_{b \rightarrow +\infty} \left(\ln \frac{b+1}{b} - \ln \frac{1+1}{1} + 1 \right) = 1 - \ln 2 + \lim_{b \rightarrow +\infty} \ln \left(1 + \frac{1}{b} \right) = 1 - \ln 2$$

2376

$$\int_0^{+\infty} x e^{-x^2} dx = \int_0^{+\infty} -\frac{1}{2} e^{-x^2} dx^2 = \lim_{b \rightarrow +\infty} -\frac{1}{2} \int_0^b e^{-x^2} dx^2 = -\frac{1}{2} \lim_{b \rightarrow +\infty} e^{-x^2} \Big|_0^b =$$

$$= -\frac{1}{2} \lim_{b \rightarrow +\infty} (e^{-b^2} - e^0) = -\frac{1}{2} \cdot (-1) = \frac{1}{2}$$

2374

$$\int_{\sqrt{2}}^{+\infty} \frac{dx}{x\sqrt{x^2-1}} = \lim_{b \rightarrow +\infty} \int_{\sqrt{2}}^b \frac{dx}{x\sqrt{x^2-1}} \quad \text{Bernoulli} \int \frac{dx}{x\sqrt{x^2-1}}$$

Заменим $t = \sqrt{x^2 - 1}$
 $\sqrt{t^2 + 1} = x$

$$dx = \frac{1}{2\sqrt{t^2 + 1}} \cdot 2t dt$$

(2)

$$\int \frac{dx}{x\sqrt{x^2 - 1}} = \int \frac{2t dt}{2\sqrt{t^2 + 1}} \cdot \frac{1}{\sqrt{t^2 + 1}} \cdot \frac{1}{t} = \int \frac{dt}{t^2 + 1} = \operatorname{arctg} t = \operatorname{arctg} \sqrt{x^2 - 1}$$

$$\begin{aligned} \Rightarrow \lim_{b \rightarrow +\infty} (\operatorname{arctg} \sqrt{x^2 - 1}) \Big|_{\sqrt{2}}^b &= \lim_{b \rightarrow +\infty} (\underbrace{\operatorname{arctg} \sqrt{b^2 - 1}}_{\rightarrow \frac{\pi}{2}} - \underbrace{\operatorname{arctg} \sqrt{2 - 1}}_{\rightarrow \frac{\pi}{4}}) = \\ &= \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4} \end{aligned}$$

2382

$$\int_1^{+\infty} \frac{\operatorname{arctg} x}{x^2} dx. \text{ Попробуем } \int \frac{\operatorname{arctg} x}{x^2} dx. = \text{II}$$

$$\begin{aligned} \text{II} &= \frac{\operatorname{arctg} x}{x^2} \cdot x - \int x d\left(\frac{\operatorname{arctg} x}{x^2}\right) = \frac{\operatorname{arctg} x}{x} - \int x \cdot \left(\frac{x^2}{1+x^2} \cdot \frac{1}{x^4} - \right. \\ &\quad \left. - \frac{2x \operatorname{arctg} x}{x^4}\right) dx = \frac{\operatorname{arctg} x}{x} - \int \left(\frac{1}{x(x^2+1)} - \frac{2 \operatorname{arctg} x}{x^2}\right) dx = \\ &= \frac{\operatorname{arctg} x}{x} - \int \frac{dx}{x(x^2+1)} + 2 \int \frac{\operatorname{arctg} x}{x^2} dx \end{aligned}$$

$$\text{II} = \frac{\operatorname{arctg} x}{x} - \int \frac{dx}{x^2(x^2+1)} + 2 \text{II}$$

$$\text{II} = \int \frac{dx}{x^2(x^2+1)} - \frac{\operatorname{arctg} x}{x}$$

12]: Заменим $x = \operatorname{tg} t$, $dx = \frac{dt}{\cos^2 t}$: $\int \frac{dx}{x(x^2+1)} = \int \frac{dt}{\cos^2 t} \cdot \frac{1}{\operatorname{tg} t}$

$$\cdot \frac{1}{(\operatorname{tg}^2 x + 1)} = \int \frac{dt}{\cos^2 t} \cdot \frac{\cos t}{\sin t} \cdot \frac{\cos^2 t}{1} = \int \operatorname{ctg} t dt = \int \frac{d(\sin t)}{\sin t} =$$

$$= \ln |\sin t| = \ln |\sin \operatorname{tg} x| = \ln |\sin(\operatorname{arctg} x)| = \ln \left| \frac{x}{\sqrt{1+x^2}} \right| + C$$

$$\text{II} = \ln \left| \frac{x}{1+x^2} \right| - \frac{\operatorname{arctg} x}{x} = \int \frac{\operatorname{arctg} x}{x^2} dx \quad (3)$$

$$\begin{aligned} \int_1^{+\infty} \frac{\operatorname{arctg} x}{x^2} dx &= \lim_{b \rightarrow +\infty} \left(\ln \left| \frac{x}{1+x^2} \right| - \frac{\operatorname{arctg} x}{x} \right) \Big|_1^b = \lim_{b \rightarrow +\infty} \left(\ln \left| \frac{b}{1+b^2} \right| - \frac{\operatorname{arctg} b}{b} - \left(\ln \left| \frac{1}{1+1} \right| - \operatorname{arctg} 1 \right) \right) \\ &= \frac{\pi}{4} + \lim_{b \rightarrow +\infty} \left(0 - \frac{\operatorname{arctg} b}{b} - \ln \frac{1}{2} \right) = \frac{\pi}{4} - \ln \frac{1}{2} = \frac{\pi}{4} - \ln 2^{-1/2} = \frac{\pi}{4} + \frac{1}{2} \ln 2 \end{aligned}$$

$$\int_3^5 \frac{x^2 dx}{(x-3)(5-x)} = \int_3^5 \frac{x^2 dx}{5x-15-x^2+3x} = \int_3^5 \frac{x^2 dx}{-x^2+8x-16+1} = \int_3^5 \frac{x^2 dx}{\sqrt{-(x-4)^2+1}}$$

Замена: $t = x - 4$
 $x = t + 4$
 $dx = dt$

$$\int_{-1}^1 \frac{(t+4)^2 dt}{\sqrt{1-t^2}} = \lim_{\varepsilon \rightarrow 0} \int_{-1-\varepsilon}^{1-\varepsilon} \frac{(t+4)^2 dt}{\sqrt{1-t^2}}$$

Бармену: $\int \frac{(t+4)^2 dt}{\sqrt{1-t^2}}$, замена $t = \sin u$, $\cos u = \sqrt{1-t^2}$
 $dt = \cos u du$
 $\sqrt{1-t^2} = \sqrt{1-\sin^2 u} = \cos u$

$$\int \frac{(\sin u + 4)^2}{\cos^2 u} \cdot \cos u du =$$

$$= \int (\sin^2 u + 8 \sin u + 16) du = \int \sin^2 u du + 8 \int \sin u du + 16 \int du =$$

$$= \frac{1}{2} \int \frac{1 - \cos 2u}{2} du + 8(-\cos u) + 16u = \frac{1}{4} u - \frac{1}{4} \sin 2u - 8 \cos u + 16u =$$

$$= \frac{33}{2} u - \frac{1}{4} \sin 2u - 8 \cos u = \frac{33}{2} u - \cos u \left(8 + \frac{1}{4} \sin u \right) =$$

$$= \frac{33}{2} \arcsin t - \sqrt{1-t^2} \left(8 + \frac{1}{2} \sin \arcsin t \right) = \frac{33}{2} \arcsin t - \sqrt{1-t^2} \cdot$$

$$\cdot \left(8 + \frac{1}{2} t \right)$$

$$\lim_{\varepsilon \rightarrow 0} \int_{-1-\varepsilon}^{1-\varepsilon} \frac{(t+4)^2 dt}{\sqrt{1-t^2}} = \lim_{\varepsilon \rightarrow 0} \left(\frac{33}{2} \arcsin t - \sqrt{1-t^2} \left(8 + \frac{1}{2} t \right) \right) \Big|_{-1-\varepsilon}^{1-\varepsilon} =$$

④

$$\lim_{\varepsilon \rightarrow 0} \left(\frac{33}{2} \arcsin(1-\varepsilon) - \frac{33}{2} \arcsin(-1-\varepsilon) \right) = \frac{33}{2} \cdot \frac{\pi}{2} + \frac{33}{2} \cdot \frac{\pi}{2} = \frac{33}{2} \cdot \pi$$

2405

$$\int_{-1}^1 \frac{dx}{(2-x)\sqrt{1-x^2}} = \left[x = \sin t, \quad dx = \cos t dt \right] = \int_{-\pi/2}^{\pi/2} \frac{\cos t dt}{(2 - \sin t)\sqrt{1 - \sin^2 t}} =$$

$$= \int_{-\pi/2}^{\pi/2} \frac{dt}{2 - \sin t} = \left[u = \tan \frac{t}{2}, \quad \sin t = \frac{2u}{1+u^2}, \quad dt = \frac{2du}{1+u^2} \right] = \int_{-1}^1 \frac{2du}{1+u^2} \cdot \frac{1}{2 - \frac{2u}{1+u^2}} =$$

	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$
$\sin$	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$
$\cos$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$
$\tan$	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$

$$= \int_{-1}^1 \frac{2du}{1+u^2} \cdot \frac{1+u^2}{2+2u^2-2u} = \int_{-1}^1 \frac{2du}{2(u^2-u+1)} =$$

$$= \int_{-1}^1 \frac{d(u-\frac{1}{2})}{(u-\frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2} = \left(\frac{2}{\sqrt{3}} \arctan \frac{2}{\sqrt{3}} \cdot (u - \frac{1}{2}) \right) \Big|_{-1}^1 =$$

$$= \frac{2}{\sqrt{3}} \left(\arctan \frac{2}{\sqrt{3}} \cdot (1 - \frac{1}{2}) - \arctan \frac{2}{\sqrt{3}} \cdot (-1 - \frac{1}{2}) \right) = \frac{2}{\sqrt{3}} \cdot \left(\arctan \frac{2}{\sqrt{3}} \cdot \frac{1}{2} + \arctan \frac{2}{\sqrt{3}} \cdot \frac{3}{2} \right) = \frac{2}{\sqrt{3}} \left(\frac{\pi}{6} + \frac{2\pi}{6} \right) = \frac{3}{\sqrt{3}} \cdot \frac{2}{\sqrt{3}} \cdot \pi = \frac{\pi}{\sqrt{3}}$$

2406

$$\int_{-1}^1 \frac{3x^2+2}{\sqrt[3]{x^2}} dx = \int_{-1}^1 \left(3 \cdot x^{2-\frac{2}{3}} + 2 \cdot x^{-\frac{2}{3}} \right) dx = \left(3 \cdot \frac{x^{7/3}}{7/3} + 2 \cdot \frac{x^{1/3}}{1/3} \right) \Big|_{-1}^1 =$$

$$= \left(\frac{9}{7} \cdot x^{7/3} + 6 \cdot x^{1/3} \right) \Big|_{-1}^1 = \left(3\sqrt[3]{x} \left(\frac{3}{7} x^2 + 2 \right) \right) \Big|_{-1}^1 = \left(\frac{9}{7} x^{7/3} + 6\sqrt[3]{x} \right) \Big|_{-1}^0 +$$

$$+ \left(\frac{9}{7} x^{7/3} + 6\sqrt[3]{x} \right) \Big|_0^{-1} = \left(\frac{9}{7} \cdot (-1) + 6 \cdot (-1) \right) + \left(\frac{9}{7} \cdot 1 + 6 \cdot 1 \right) = \frac{102}{7} = 14 \frac{4}{7}$$

2409

$$\frac{1}{2} = \int_{-1}^1 \frac{\ln(2+\sqrt[3]{x})}{\sqrt[3]{x}} dx = \left[t^3 = x \quad dt = 3t^2 dt \right] =$$

$$= \int_{-1}^1 \frac{\ln(2+t)}{t} \cdot 3t^2 dt = 3 \int_{-1}^1 \ln(t+2) t dt = \frac{3}{2} \int \ln(t+2) dt^2 =$$

$$= \frac{3}{2} (\ln(t+2) \cdot t^2 - \int t^2 \cdot d(\ln(t+2))) = \frac{3}{2} t^2 \ln(t+2) -$$

$$- \frac{3}{2} \int \frac{t^2}{t+2} dt = \frac{3}{2} t^2 \ln(t+2) - \frac{3}{2} \int ((t-2) + \frac{4}{t+2}) dt = \frac{3}{2} t^2 \ln(t+2) -$$

$$- \frac{3}{2} \int (t-2) d(t-2) - \frac{3}{2} \cdot 4 \int \frac{d(t+2)}{t+2} = \frac{3}{2} t^2 \ln(t+2) - \frac{3}{2} \frac{(t-2)^2}{2} -$$

$$- \frac{3}{2} \cdot 4 \ln(t+2) = \frac{3}{2} \left(\frac{(t-2)^2}{-2} + \ln(t+2)(t^2-4) \right)$$

$$\int_{-1}^1 \frac{\ln(2+\sqrt[3]{x})}{\sqrt[3]{x}} dx = \frac{3}{4} (\sqrt[3]{x}-2)^2 + \frac{3}{2} \ln(\sqrt[3]{x}+2)(\sqrt[3]{x^2}-4) \Big|_{-1}^1 =$$

$$= \left[\frac{3}{4} (\sqrt[3]{x}-2)^2 + \frac{3}{2} \ln(\sqrt[3]{x}+2)(\sqrt[3]{x^2}-4) \right] \Big|_{-1}^0 + \left[\frac{3}{4} (\sqrt[3]{x}-2)^2 + \right.$$

$$\left. + \frac{3}{2} \ln(\sqrt[3]{x}+2)(\sqrt[3]{x^2}-4) \right] \Big|_0^1 = \left[\frac{3}{4} (0-2)^2 + \frac{3}{2} \ln(0+2)(0-4) - \right.$$

$$\left. - \frac{3}{4} (-1-2)^2 - \frac{3}{2} \ln(-1+2)(-1-4) \right] + \left[\frac{3}{4} (1-2)^2 + \frac{3}{2} \ln(1+2)(1-4) - \right.$$

$$\left. - \frac{3}{4} (0-2)^2 - \frac{3}{2} \ln(0+2)(0-4) \right] = 3 - 6 \ln 2 - \frac{27}{4} - 0 +$$

$$+ \frac{3}{4} + \frac{3}{2} \cdot (3) \ln 3 - 3 + 6 \ln 2 = \frac{24}{4} - 3 \cdot \frac{3}{2} \ln 3 = 6 - \frac{9}{2} \ln 3$$

$$\int_0^1 \frac{dx}{1-x^2+2\sqrt{1-x^2}} = \lim_{b \rightarrow 1} \int_0^b \frac{dx}{1-x^2+2\sqrt{1-x^2}} \quad (2404)$$

$$\int \frac{dx}{1-x^2+2\sqrt{1-x^2}} = \left[\begin{array}{l} x = \sin t \quad dx = \cos t dt \\ t = \arcsin x \quad 1-x^2 = \cos^2 t \end{array} \right] = \int \frac{\cos t dt}{\cos^2 t + 2 \cos t} =$$

$$x_1 = b \rightarrow 1 \Rightarrow t_1 \rightarrow \frac{\pi}{2}$$

$$x_0 = 0 \Rightarrow t_0 = 0$$

$$\textcircled{6} \quad \int \frac{dt}{\cos t + 2} = \int u = \operatorname{tg} \frac{t}{2} \quad \left[\begin{array}{l} \cos t = \frac{1-u^2}{1+u^2} \\ dt = \frac{2du}{1+u^2} \\ t \rightarrow \pi/2 \Rightarrow u_1 \rightarrow 1 \\ t_0 = 0 \Rightarrow u_0 = 0 \end{array} \right] = \int \frac{2du}{1+u^2}.$$

$$\cdot \frac{1}{\frac{1-u^2}{1+u^2} + 2} = \int \frac{2du}{1+u^2} \cdot \frac{1+u^2}{1-u^2+2+2u^2} = \int \frac{2du}{3+u^2} = \frac{2}{\sqrt{3}} \cdot \operatorname{arctg} \frac{u}{\sqrt{3}}$$

$$\Rightarrow \lim_{u_1 \rightarrow 1} \int_0^{u_1} \frac{2du}{3+u^2} = \lim_{u_1 \rightarrow 1} \frac{2}{\sqrt{3}} \operatorname{arctg} \frac{u}{\sqrt{3}} \Big|_0^{u_1} = \lim_{u_1 \rightarrow 1} \frac{2}{\sqrt{3}} \left(\operatorname{arctg} \frac{u_1}{\sqrt{3}} - \operatorname{arctg} 0 \right) = \frac{\pi}{6} \cdot \frac{2}{\sqrt{3}} = \frac{\pi}{3\sqrt{3}}$$